

MUHAMMAD SULAIMAN, PhD

Professor of Mathematics | Applied & Computational Mathematics | Optimization | Scientific Machine Learning & PINNs

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[Google Scholar](#) | [ORCID](#) | [Scopus](#) | [Web of Science](#) | [LinkedIn](#) | [ResearchGate](#)

ACADEMIC PROFILE

Applied and computational mathematician and tenured Full Professor with 16+ years of teaching and research experience. Research spans global and multi-objective optimization, evolutionary computation, scientific machine learning, physics-informed neural networks (PINNs), numerical differential equations, and mathematical modelling. The source CV records 105 publications (103 journal articles and 2 book chapters), including 30 Q1, 48 Q2, and 6 Q3 journal papers; Google Scholar h-index 30 with 2,693 citations as of 12 July 2026. Principal supervisor of 8 completed PhD and 40 completed MPhil degrees, with extensive international collaboration, grant participation, teaching, reviewing, and academic service.

Scientific ML & Physics-Informed Neural Networks	Global, Multi-objective & Engineering Optimization	Evolutionary Computation & Metaheuristics
Numerical ODEs, PDEs & Fractional Differential Equations	Mathematical Modelling of Physical, Biological & Cyber Systems	Fluid Dynamics, Heat/Mass Transfer & Computational Intelligence

RESEARCH IMPACT AT A GLANCE

105 Publications	30 h-index	2,693 Google Scholar citations	8 PhD / 40 MPhil Completed as principal supervisor
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JCR quartile profile: 30 Q1 | 48 Q2 | 6 Q3 | 0 Q4 • 2 book chapters • cumulative impact factor 285+

ACADEMIC APPOINTMENTS

2025-Present	Full Professor (Tenured) Department of Mathematics, Faculty of Physical and Numerical Sciences, Abdul Wali Khan University Mardan, Pakistan
2021-2025	Associate Professor (Tenured) Abdul Wali Khan University Mardan, Pakistan
2016-2021	Assistant Professor (Tenure Track) Abdul Wali Khan University Mardan, Pakistan
2009-2016	Lecturer (BPS-18) Abdul Wali Khan University Mardan, Pakistan

EDUCATION

2011-2015	PhD in Mathematics - University of Essex, United Kingdom Thesis: A Nature-Inspired Metaheuristic: The Plant Propagation Algorithm. Passed with no corrections.
2007-2009	MPhil in Mathematics - Quaid-i-Azam University, Islamabad, Pakistan Thesis: Extension of Noether's theorem to constrained non-conservative dynamical systems in Poincare formalism. First Division, 77%.
2005-2007	MSc in Mathematics - Quaid-i-Azam University, Islamabad, Pakistan First Division, 70%.
2002-2004	BSc - University of Peshawar, Pakistan Mathematics-A, Mathematics-B, Computer Science. First Division, 72%.

RESEARCH GRANTS & FUNDED PROJECTS

2021	Principal Investigator - Application of Soft Computing Techniques for Solving Real-life Problems International collaboration with Universidad Santiago de Cali, Colombia. Project amount: USD 26,008.
2016	Principal Investigator - Startup Research Grant Higher Education Commission Pakistan. Amount: PKR 0.2 million.
2022	Research grant - King Faisal University, Saudi Arabia Deanship of Scientific Research / Vice Presidency for Graduate Studies and Scientific Research. Amount: USD 6,000.
2022-2025	Foreign Researcher - Fundamental Research Grant Scheme (FRGS), Malaysia Project: A new neuro soft-computing based on green former matrix factorization of high dimensional partial differential equations for ecological modelling. Amount: USD 15,900.
2022-2023	Supervisor - Undergraduate Research Support Program Grant Directorate General of Science & Technology, Khyber Pakhtunkhwa. Project on a machine-learning framework for a COVID-19 model. Amount: PKR 150,000.

GRADUATE SUPERVISION & EXAMINATION

- 8 PhD graduates supervised to completion as principal supervisor.
- 40 MPhil graduates supervised to completion as principal supervisor.
- 38 BS Mathematics research projects supervised.
- External examiner for BS/MPhil/PhD work at the University of Malakand, HITEC University Taxila, Kohat University of Science and Technology, Shaheed Benazir Bhutto Women University, University of Swat, UET Peshawar, and COMSATS University Islamabad.

TEACHING PORTFOLIO

Optimization, AI & computational methods: Bio-inspired Algorithms for Optimization; Advanced Metaheuristics; Evolutionary Algorithms and Applications; Neural Networks and Optimization Algorithms; Optimization Theory and Computational Methods; Engineering Optimization; Introduction to Artificial Intelligence; Programming for Artificial Intelligence.

Core mathematics: Ordinary Differential Equations; Partial Differential Equations; Calculus I-II; Linear Algebra I; Real Analysis; Basic Mathematics; Introduction to Optimization Theory.

HONORS, RECOGNITION & PROFESSIONAL STANDING

- Faculty Development Scholarship, Government of Khyber Pakhtunkhwa, Pakistan, 2011.
- Silver Medal for Best Paper in Engineering Design, Innovation & Technology Transformation Post-COVID-19, Universiti Sultan Zainal Abidin (UniSZA), Malaysia, 2022.
- Associate Fellow, Higher Education Academy, United Kingdom, 2014.
- Higher Education Commission Pakistan Approved PhD Supervisor, 2015-present.

TECHNICAL & LANGUAGE SKILLS

Scientific/technical: MATLAB, Python, LaTeX, Mathematica, Maple, OriginLab; MS Office; online teaching/collaboration tools including Zoom and MS Teams.

Languages: English (advanced), Urdu (advanced), Pashto (advanced).

ACADEMIC SERVICE & PROFESSIONAL CONTRIBUTIONS

- Reviewer for 26 journals/books, including Neural Computing and Applications, IEEE Access, IEEE Transactions on Systems, Man, and Cybernetics: Systems, PLOS ONE, Journal of Computational Design and Engineering, Fractal and Fractional, Soft Computing, RAIRO - Operations Research, and Journal of the Operational Research Society.
- Editorial board and special-issue roles across 9 outlets, including Energies, Computers and Electrical Engineering, IEEE Journal of Biomedical and Health Informatics, Fuzzy Information and Engineering, and Computing and Informatics.
- Program/technical/scientific committee service for 5 international conferences, including the 14th Metaheuristics International Conference (2022), ICNC-FSKD, and ICADET.
- University service at Abdul Wali Khan University Mardan includes Board of Studies, Graduate Studies Committee, Departmental Tenure Review Committee, Scholarship Committee, Seminar Organizing Committee, Teacher Evaluation Team, and curriculum/syllabi review committees.
- Participation in 14 regional and international conferences/workshops from 2012 to 2024, including META 2014, MESS 2018, NUST conferences, and the 2024 International Conference on Multidisciplinary Design Optimization and Applications.

SELECTED PUBLICATIONS - SCIENTIFIC ML, PINNS & OPTIMIZATION

- Alkathlan, Ali; Kateb, Faris; Ahmad, Tauqeer; **Sulaiman, Muhammad**; Salman, Mohammed Abdullah. "Cuckoo Search Optimized Fractional Physics-Informed Neural Networks for Fourth-Order Time-Fractional Partial Differential Equations." Journal of Mathematics, 2026.
- Fazal, Fazl Ullah; **Sulaiman, Muhammad**; Ahmed Ali, Mustafa. "Numerical Simulation of Generalized Fractional Thermoelasticity with Two Temperature and Phase-Lags in a Half-Space using GLA-PINN." Complexity, Vol. 2026, 2026.
- Saad, Muhammad; **Sulaiman, Muhammad**; Ur Rahman, Izaz; Alshammari, Fahad Sameer; Laouini, Ghaylen; Abdullah Salman, Mohammed. "Learning PINN-Based Coupled Fractional Systems with Memory and Nonlocal Dynamics Using Neural Computing." Journal of Mathematics, Vol. 2026, 2026.
- Khan, Hafsa; **Sulaiman, Muhammad**; Laouini, Ghaylen; Ur Rahman, Izaz; Salman Abdullah, Mohammed. "Physics-Informed Neural Networks with Caputo-Fabrizio Derivatives for Nonlinear Fractal-Fractional Delay Equations and Chaotic Systems." Scientific Reports, 2026.
- Jan, Hamad; **Sulaiman, Muhammad**; Ahmad, Tauqeer; Laouini, Ghaylen; Rahman, Izaz Ur; Abdullah Salman, Mohammed. "Computational modeling of Monte Carlo-enhanced PINNs for fractional order differential models with memory effects." Scientific Reports, 2026.
- Ahmad, Tauqeer; **Sulaiman, Muhammad**; Bassir, David; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Enhanced Numerical Solutions for Fractional PDEs Using Monte Carlo PINNs Coupled with Cuckoo Search Optimization." Fractal and Fractional, 2025.
- Fazal, Fazlullah; **Sulaiman, Muhammad**; Bassir, David; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Quantitative Analysis of the Fractional Fokker-Planck Levy Equation via a Modified Physics-Informed Neural Network Architecture." Fractal and Fractional, 2024.
- Khan, Muhammad Fawad; Hu, Gang; **Sulaiman, Muhammad**. "Transformer model learning driven analysis of Casson fluid flow influenced by Hall currents and Darcy-Forchheimer effects." Physics of Fluids, Vol. 37, No. 6, 2025.
- Sulaiman, Muhammad**; Waseem, Muhammad; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer. "Defense strategies for epidemic cyber security threats: modeling and analysis by using a machine learning approach." IEEE Access, Vol. 12, pp. 4958-4984, 2024.
- Sulaiman, Muhammad**; Khalaf, Osamah Ibrahim; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Hamam, Habib. "Mathematical modeling and machine learning-based optimization for enhancing biofiltration efficiency of volatile organic compounds." Scientific Reports, Vol. 14, No. 1, Article 16908, 2024.
- Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Lu, Benzhou. "ANN based optimization of nano-beam oscillations with intermolecular forces and geometric nonlinearity." International Journal of Solids and Structures, Vol. 304, Article 113054, 2024.
- Sulaiman, Muhammad**; Khan, Awais; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer. "Quantitative analysis of worm transmission and insider risks in air-gapped networking using a novel machine learning approach." IEEE Access, 2023.
- Sulaiman, Muhammad**; Ahmad, Sohail; Iqbal, Javed; Khan, Asfandiyar; Khan, Rahim. "Optimal operation of the hybrid electricity generation system using multiverse optimization algorithm." Computational Intelligence and Neuroscience, Vol. 2019, No. 1, Article 6192980, 2019.
- Sulaiman, Muhammad**; Salhi, Abdellah; Selamoglu, Birsan Irem; Kirikchi, Omar Bahaaldin. "A plant propagation algorithm for constrained engineering optimisation problems." Mathematical Problems in Engineering, Vol. 2014, No. 1, Article 627416, 2014.

TRAINING

- University of Essex, UK - Induction Program for Graduate Teaching Assistants.
- University of Essex, UK - Equality & Diversity Essentials, score 88%.

Complete publication list follows. Online publication profile: [Google Scholar](#)

COMPLETE PUBLICATION LIST

2026

- Rahman, Irfan Ur; **Sulaiman, Muhammad**; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Numerical study of nonlinear flow and mass transfer through a channel filled with a porous medium using a hybrid heuristic paradigm." *Waves in Random and Complex Media*, Vol. 36, No. 2, pp. 1448-1483, 2026.
- Sulaiman, Muhammad**; Fazal, Fazlullah; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Dynamics of bioconvection radiative MHD flow with gyrotactic microorganisms: an intelligent computation approach." *Waves in Random and Complex Media*, Vol. 36, No. 3, pp. 3874-3906, 2026.
- Sulaiman, Muhammad**; Hussain, Zaware; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Computational Analysis of Natural Convection Heat Transfer in Nanofluids Under a Uniform Magnetic Field Using Levenberg-Marquardt Backpropagation Neural Networks." *Journal of Computational Applied Mechanics*, Vol. 57, No. 1, pp. 41-62, 2026.
- Saad, Muhammad; **Sulaiman, Muhammad**; Khan, Muhammad Fawad; Laouini, Ghaylen; Ashraf, Rashid; Alshammari, Fahad Sameer. "Magnetic Hydrodynamic Flow and Heat Transfer of Williamson Nanofluids in a Porous Medium Impact of Chemical Reactions and Melting Effects." *Journal of Computational Applied Mechanics*, Vol. 57, No. 2, pp. 173-194, 2026.
- Jan, Hamad; **Sulaiman, Muhammad**; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Laouini, Ghaylen; Ashraf, Rashid; Ali, and Mustafa Ahmed. "Improved pipeline damage detection using flexural and torsional ultrasonic guided waves." *Research in Mathematics*, Vol. 13, No. 1, 2026.
- Alkathlan, Ali; Kateb, Faris; Ahmad, Tauqeer; **Sulaiman, Muhammad**; Salman, Mohammed Abdullah. "Cuckoo Search Optimized Fractional Physics-Informed Neural Networks for Fourth-Order Time-Fractional Partial Differential Equations." *Journal of Mathematics*, 2026.
- Saad, Muhammad; **Sulaiman, Muhammad**; Khan, Naveed Ahmad; Laouini, Ghaylen; Alshammari, Fahad Sameer; Rahman, Izaz Ur; Ali, and Mustafa Ahmed. "Neurocomputing Modelling of Hydromagnetic Nanofluid Flow with Chemical Reactions on Fluctuating Sheets." *Computational and Mathematical Methods*, 2026.
- Fazal, Fazl Ullah; **Sulaiman, Muhammad**; Ahmed Ali, Mustafa. "Numerical Simulation of Generalized Fractional Thermoelasticity with Two Temperature and Phase-Lags in a Half-Space using GLA-PINN." *Complexity*, Vol. 2026, 2026.
- Saad, Muhammad; **Sulaiman, Muhammad**; Ur Rahman, Izaz; Alshammari, Fahad Sameer; Laouini, Ghaylen; Abdullah Salman, Mohammed. "Learning PINN-Based Coupled Fractional Systems with Memory and Nonlocal Dynamics Using Neural Computing." *Journal of Mathematics*, Vol. 2026, 2026.
- Jan, Hamad; **Sulaiman, Muhammad**; Fawad Khan, Muhammad; Laouini, Ghaylen; Rahman, Izaz Ur; Abdullah Salman, Mohammed. "Predictive Modeling of Pest Spread in Tea Plants Using an Intelligent Computational Approach." *Scientific Reports*, 2026.
- Khan, Hafsa; **Sulaiman, Muhammad**; Laouini, Ghaylen; Ur Rahman, Izaz; Salman Abdullah, Mohammed. "Physics-Informed Neural Networks with Caputo-Fabrizio Derivatives for Nonlinear Fractal-Fractional Delay Equations and Chaotic Systems." *Scientific Reports*, 2026.
- Jan, Hamad; **Sulaiman, Muhammad**; Ahmad, Tauqeer; Laouini, Ghaylen; Rahman, Izaz Ur; Abdullah Salman, Mohammed. "Computational modeling of Monte Carlo-enhanced PINNs for fractional order differential models with memory effects." *Scientific Reports*, 2026.
- Saad, Muhammad; **Sulaiman, Muhammad**; Alshammari, Fahad Sameer; Rahman, Izaz Ur; Salman, Mohammed Abdullah. "Unveiling Hidden Patterns in Energy Consumption with a Multi-Level Machine Learning Framework." *Research in Mathematics*, 2026.

2025

- Ahmad, Tauqeer; **Sulaiman, Muhammad**; Bassir, David; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Enhanced Numerical Solutions for Fractional PDEs Using Monte Carlo PINNs Coupled with Cuckoo Search Optimization." *Fractal and Fractional*, 2025.
- Khan, Muhammad Fawad; Hu, Gang; **Sulaiman, Muhammad**. "Transformer model learning driven analysis of Casson fluid flow influenced by Hall currents and Darcy-Forchheimer effects." *Physics of Fluids*, Vol. 37, No. 6, 2025.
- Jan, Hamad; **Sulaiman, Muhammad**; Saad, Muhammad; Khan, Awais; Salman, Muhammad. "Impact of Individual Participation and Their Health Education on the Growth of the Sportsy Population." *VFAST Transactions on Mathematics*, Vol. 13, No. 1, pp. 59-83, 2025.
- Khan, A.; Ahmad, T.; Khan, N. A.; **Sulaiman, Muhammad**. "Predictive Modeling of Parameter Variations in Conveyor Belt Dryers using Machine Learning for Improved Drying Performance." *VFAST Transactions on Mathematics*, Vol. 13, No. 2, pp. 01-19, 2025.

2024

- Sulaiman, Muhammad**; Waseem, Muhammad; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer. "Defense strategies for epidemic cyber security threats: modeling and analysis by using a machine learning approach." *IEEE Access*, Vol. 12, pp. 4958-4984, 2024.
- Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Lu, Benzhou. "Predictive insights into nonlinear nanofluid flow in rotating systems: a machine learning approach." *Engineering with Computers*, 2024.
- Sulaiman, Muhammad**; Khalaf, Osamah Ibrahim; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Algburi, Sameer; Hamam, Habib. "Machine Learning-Based Prediction of Heat Transfer Performance in Annular Fins with Functionally Graded Materials." *Scientific Reports*, 2024.
- Sulaiman, Muhammad**; Salman, Muhammad; Laouini, Ghaylen; Alshammari, Fahad Sameer. "Mathematical models for fluid flow in porous media with machine learning techniques for landfill waste leachate." *Stochastic Environmental Research and Risk Assessment*, 2024.
- Sulaiman, Muhammad**; Khalaf, Osamah Ibrahim; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Hamam, Habib. "Mathematical modeling and machine learning-based optimization for enhancing biofiltration efficiency of volatile organic compounds." *Scientific Reports*, Vol. 14, No. 1, Article 16908, 2024.
- Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Lu, Benzhou. "ANN based optimization of nano-beam oscillations with intermolecular forces and geometric nonlinearity." *International Journal of Solids and Structures*, Vol. 304, Article 113054, 2024.
- Fazal, Fazlullah; **Sulaiman, Muhammad**; Bassir, David; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Quantitative Analysis of the Fractional Fokker-Planck Levy Equation via a Modified Physics-Informed Neural Network Architecture." *Fractal and Fractional*, 2024.

2023

- Alshammari, Fahad Sameer; Jan, Hamad; **Sulaiman, Muhammad**; Prathumwan, Din; Laouini, Ghaylen. "Quantitative study of non-linear convection diffusion equations for a rotating-disc electrode." *Entropy*, Vol. 25, No. 1, Article 134, 2023.
- Sulaiman, Muhammad**; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Performance of heat transfer in micropolar fluid with isothermal and isoflux boundary conditions using supervised neural networks." *Mathematics*, Vol. 11, No. 5, Article 1173, 2023.

- Khan, Muhammad Fawad; **Sulaiman, Muhammad**; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer; Khalid, Majdi. "A computational study of magneto-convective heat transfer over inclined surfaces with thermodiffusion." *IEEE Access*, Vol. 11, pp. 57046-57070, 2023.
- Sulaiman, Muhammad**; Khan, Naveed Ahmad. "Predictive modeling of oil and water saturation during secondary recovery with supervised learning." *Physics of Fluids*, Vol. 35, No. 6, 2023.
- Sulaiman, Muhammad**; Fazal, Fazlullah; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer; Khalid, Majdi. "A stochastic NARX neural network to investigate the carbon capture in the plantations of forests." *IEEE Access*, Vol. 11, pp. 74702-74721, 2023.
- Sulaiman, Muhammad**; Khan, Awais; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer. "Quantitative analysis of worm transmission and insider risks in air-gapped networking using a novel machine learning approach." *IEEE Access*, 2023.

2022

- Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Jeza Aljohani, Abdulah; Abu Bakar, Maharani; Miftahuddin, Miftahuddin. "Mathematical models of CBSC over wireless channels and their analysis by using the LeNN-WOA-NM algorithm." *Engineering Applications of Artificial Intelligence*, 2022.
- Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Tavera Romero, Carlos Andrés; Laouini, Ghaylen; Alshammari, Fahad Sameer. "Study of rolling motion of ships in random beam seas with nonlinear restoring moment and damping effects using neuroevolutionary technique." *Materials*, Vol. 15, No. 2, Article 674, 2022.
- Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Kumam, Poom; Alarfaj, Fawaz Khaled. "Application of Legendre polynomials based neural networks for the analysis of heat and mass transfer of a non-Newtonian fluid in a porous channel." *Advances in Continuous and Discrete Models*, Vol. 2022, No. 1, pp. 1-32, 2022.
- Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Tavera Romero, Carlos Andrés; Alshammari, Fahad Sameer. "Analysis of Nanofluid Particles in a Duct with Thermal Radiation by Using an Efficient Metaheuristic-Driven Approach." *Nanomaterials*, 2022.
- Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Bonya, Ebenezer; Seidu, Jamel; Alshammari, Fahad Sameer. "Investigation of three-dimensional condensation film problem over an inclined rotating disk by using nonlinear autoregressive exogenous model." *Computational Intelligence and Neuroscience*, 2022.
- Ganie, Abdul Hamid; Fazal, Fazlullah; Tavera Romer, Carlos Andrés; **Sulaiman, Muhammad**. "Quantitative features analysis of water carrying nanoparticles of alumina over a uniform surface." *Nanomaterials*, Vol. 12, No. 5, Article 878, 2022.
- Guo, Jiayi; Khan, Adnan; **Sulaiman, Muhammad**; Kumam, Poom. "A novel neuroevolutionary paradigm for solving strongly nonlinear singular boundary value problems in physiology." *IEEE Access*, Vol. 10, pp. 21979-22002, 2022.
- Khan, Muhammad Fawad; **Sulaiman, Muhammad**; Romero, Carlos Andres Tavera; Alshammari, Fahad Sameer. "A quantitative study of non-linear convective heat transfer model by novel hybrid heuristic driven neural soft computing." *IEEE Access*, Vol. 10, pp. 34133-34153, 2022.
- Sulaiman, Muhammad**; Umar, Muhammad; Nonlaopon, Kamsing; Sameer Alshammari, Fahad. "The Quantitative Features Analysis of the Nonlinear Model of Crop Production by Hybrid Soft Computing Paradigm." *Agronomy*, 2022.
- Ganie, Abdul Hamid; Rahman, Irfan Ur; **Sulaiman, Muhammad**; Nonlaopon, Kamsing. "Solution of nonlinear reaction-diffusion model in porous catalysts arising in micro-vessel and soft tissue using a metaheuristic." *IEEE Access*, Vol. 10, pp. 41813-41827, 2022.
- Nonlaopon, Kamsing; Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Heat Transfer Analysis of Nanofluid Flow in a Rotating System with Magnetic Field Using an Intelligent Strength Stochastic-Driven Approach." *Nanomaterials*, 2022.
- Ahmad Khan, Naveed; **Sulaiman, Muhammad**. "Heat transfer and thermal conductivity of magneto micropolar fluid with thermal non-equilibrium condition passing through the vertical porous medium." *Waves in Random and Complex Media*, 2022.
- Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Sameer Alshammari, Fahad. "Heat transfer analysis of an inclined longitudinal porous fin of trapezoidal, rectangular and dovetail profiles using Cascade neural networks." *Structural and Multidisciplinary Optimization*, 2022.
- Khan, Muhammad Fawad; **Sulaiman, Muhammad**; Sameer Alshammari, Fahad. "A Hybrid Heuristic-Driven Technique to Study the Dynamics of Savanna Ecosystem." *Stochastic Environmental Research and Risk Assessment*, 2022.
- Ahmad Khan, Naveed; **Sulaiman, Muhammad**. "Investigation of nonlinear vibrational analysis of circular sector oscillator by using cascade learning." *Advances in Materials Science and Engineering*, Vol. 2022, 2022.
- Kamal, Mustafa; **Sulaiman, Muhammad**; Alshammari, Fahad Sameer. "Quantitative features analysis of a model for separation of dissolved substances from a fluid flow by using a hybrid heuristic." *The European Physical Journal Plus*, 2022.
- Alhakami, Hosam; Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Alhakami, Wajdi; Baz, Abdullah. "On the Computational Study of a Fully Wetted Longitudinal Porous Heat Exchanger Using a Machine Learning Approach." *Entropy*, 2022.
- Alhakami, Hosam; Kamal, Mustafa; **Sulaiman, Muhammad**; Alhakami, Wajdi; Baz, Abdullah. "A machine learning strategy for the quantitative analysis of the global warming impact on marine ecosystems." *Symmetry*, 2022.
- Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Seidu, Jamel; Alshammari, Fahad Sameer. "Mathematical analysis of prey predator system with immigrant prey using the soft computing technique." *Discrete Dynamics in Nature and Society*, 2022.
- Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Alshammari, Fahad Sameer. "Analysis of heat transmission in convective, radiative and moving rod with thermal conductivity by using meta-heuristic driven soft computing technique." *Structural and Multidisciplinary Optimization*, 2022.
- Nonlaopon, Kamsing; Khan, Muhammad Fawad; **Sulaiman, Muhammad**; Alshammari, Fahad Sameer; Laouini, Ghaylen. "Analysis of MHD Falkner-Skan boundary layer flow and heat transfer due to symmetric dynamic wedge: A numerical study via the SCA-SQP-ANN technique." *Symmetry*, Vol. 14, No. 10, Article 2180, 2022.
- Alhakami, Hosam; Umar, Muhammad; **Sulaiman, Muhammad**; Alhakami, Wajdi; Baz, Abdullah. "A numerical study of the dynamics of vector-born viral plant disorders using a hybrid artificial neural network approach." *Entropy*, Vol. 24, No. 11, Article 1511, 2022.
- Khaled Alarfa, Fawaz J.; Ahmad Khan, Naveed; **Sulaiman, Muhammad**; M. Alomair, Abdullah. "Application of a machine learning algorithm for evaluation of stiff fractional modeling of polytropic gas spheres and electric circuits." *Symmetry*, 2022.

2021

- Ahmad, Ashfaq; **Sulaiman, Muhammad**; Aljohani, Abdulah Jeza; Alhindi, Ahmad; Alrabaiah, Hussam. "Design of an efficient algorithm for solution of Bratu differential equations." *Ain Shams Engineering Journal*, Vol. 12, No. 2, pp. 2211-2225, 2021.
- Huang, Wen; Jiang, Tianhua; Zhang, Xiucheng; Khan, Naveed Ahmad; **Sulaiman, Muhammad**. "Analysis of Beam-Column Designs by Varying Axial Load with Internal Forces and Bending Rigidity Using a New Soft Computing Technique." *Complexity*, Vol. 2021, No. 1, Article 6639032, 2021.
- Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Kumam, Poom; Aljohani, Abdulah Jeza. "A new soft computing approach for studying the wire coating dynamics with Oldroyd 8-constant fluid." *Physics of Fluids*, Vol. 33, No. 3, 2021.

- Zhang, Yin; Lin, Jianqiang; Hu, Zhenhuan; Khan, Naveed Ahmad; **Sulaiman, Muhammad**. "Analysis of third-order nonlinear multi-singular Emden-Fowler equation by using the LeNN-WOA-NM algorithm." *IEEE Access*, Vol. 9, pp. 72111-72138, 2021.
- Khan, Naveed Ahmad; Khalaf, Osamah Ibrahim; Romero, Carlos Andrés Tavera; **Sulaiman, Muhammad**; Bakar, Maharani A. "Application of Euler Neural Networks with Soft Computing Paradigm to Solve Nonlinear Problems Arising in Heat Transfer." *Entropy*, 2021.
- Ahmad, Ashfaq; **Sulaiman, Muhammad**; Kumam, Poom. "Solutions of fractional order differential equations modeling temperature distribution in convective straight fins design." *Advances in Difference Equations*, 2021.
- Ahmad, Ashfaq; **Sulaiman, Muhammad**; Kumam, Poom; A. Bakar, Maharani; Miftahuddin. "Analysis of a Mathematical Model for Drilling System With Reverse Air Circulation by Using the ANN-BHCS Technique." *IEEE Access*, Vol. 9, pp. 118-218, 2021.
- Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Kumam, Poom; Bakar, Maharani Abu. "Thermal analysis of conductive-convective-radiative heat exchangers with temperature dependent thermal conductivity." *IEEE Access*, Vol. 9, pp. 138876-138902, 2021.
- Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Tavera Romero, Carlos Andrés; Alarfaj, Fawaz Khaled. "Theoretical analysis on absorption of carbon dioxide (CO₂) into solutions of phenyl glycidyl ether (PGE) using nonlinear autoregressive exogenous neural networks." *Molecules*, Vol. 26, No. 19, Article 6041, 2021.
- Khan, Muhammad Fawad; **Sulaiman, Muhammad**; Tavera Romero, Carlos Andrés; Alkhathlan, Ali. "Falkner-Skan flow with stream-wise pressure gradient and transfer of mass over a dynamic wall." *Entropy*, Vol. 23, No. 11, Article 1448, 2021.
- Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Tavera Romero, Carlos Andrés; Khaled Alarfaj, Fawaz. "Numerical analysis of electrohydrodynamic flow in circular cylindrical conduit by using neuro evolutionary technique." *Energies*, 2021.
- Fawad Khan, Muhammad; **Sulaiman, Muhammad**; Tavera Romero, Carlos Andrés; Alkhathlan, Ali. "A Hybrid Meta-Heuristic Based on Neuro-Computing for Analysis of Unipolar Electro-Hydrodynamic Pump Flow." *Entropy*, 2021.
- Cui, Yanmei; Hong, Yong; Ahmad Khan, Naveed; **Sulaiman, Muhammad**. "Application of soft computing paradigm to large deformation analysis of cantilever beam under point load." *Complexity*, 2021.
- Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Tavera Romero, Carlos Andrés; **Sulaiman, Muhammad**; Mirjalili, Seyedali. "An optimistic solver for the mathematical model of the flow of Johnson Segalman fluid on the surface of an infinitely long vertical cylinder." *Materials*, Vol. 14, No. 24, Article 7798, 2021.
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